Adaptive versus Fixed Thresholds for Memory Filtering in MA-SAM2-Based Video Object Segmentation

AIST4010 Project Milestone Report | Author: LAU Yat Laam Jeffery | Student ID: 1155214617

1. Introduction

Video object segmentation identifies the pixels of selected objects throughout a video. Given object masks in the first frame, the model should follow the same objects in later frames despite movement, changes in appearance, and occlusion. This requires both accurate object boundaries and consistent object identity over time.

SAM 2 processes videos using a streaming memory that stores information from earlier frames. MA-SAM2 extends this approach with Context-Aware Memory (CAM) and Occlusion-Resilient Memory (ORM) for surgical videos. These methods show the importance of managing past information rather than treating all previous predictions as equally useful.

This project studies the predicted-IoU threshold used in CAM memory filtering. Predicted IoU is the model's estimate of mask quality, not an IoU calculated from the ground-truth test mask. A strict threshold may reject useful memory updates, while a loose threshold may admit incorrect predictions. It is therefore worth investigating whether a threshold that changes with recent prediction quality works better than a fixed value.

The research question is: can an adaptive CAM predicted-IoU threshold improve segmentation compared with a development-tuned fixed threshold, using the same model and memory capacity? The planned contributions are a fixed-threshold sensitivity study, a bounded adaptive rule based on past model outputs, and an analysis of when adaptation helps or fails. The project does not assume that adaptation will always improve performance.

2. Related Work

Memory-based video object segmentation

Oh et al.'s Space-Time Memory Networks (STM, ICCV 2019) use information from previous frames and masks to segment a query frame. This established an important memory-based approach to video object segmentation. Cheng et al.'s STCN (NeurIPS 2021) studies improved memory coverage for efficient segmentation. Together, these works motivate examining which past information should remain available to the model. Cheng and Schwing's XMem (ECCV 2022) uses sensory, working, and long-term memory stores to handle different time scales. It addresses the difficulty of retaining useful history without allowing memory cost to grow continuously. The present project studies a narrower question: which predictions should pass an admission threshold when memory capacity is held constant.

Yang et al.'s AOT (NeurIPS 2021) introduces an identification mechanism and a Long Short-Term Transformer for associating multiple objects. Yang and Yang's DeAOT (NeurIPS 2022) separates object-agnostic and object-specific feature propagation. These approaches improve how object information is represented and propagated, whereas this project changes a memory-admission rule rather than the underlying representation. Cheng et al.'s Cutie (CVPR 2024) uses object-level memory reading to reduce problems caused by noisy pixel matching and similar-looking distractors. This highlights a limitation relevant to the present study: a memory entry may appear useful but still support the wrong object. Threshold analysis should therefore include qualitative examples, not only average scores.

Promptable segmentation and memory filtering

Kirillov et al.'s Segment Anything (SAM, ICCV 2023) introduces a promptable image segmentation model. Ravi et al.'s SAM 2 (ICLR 2025) extends promptable segmentation to videos through a streaming-memory architecture. SAM 2 provides the pretrained segmentation model used as the foundation of this project.

Yin et al.'s MA-SAM2 (MICCAI 2025) adds context-aware and occlusion-resilient memory handling without additional training. It is the closest related work because the proposed method changes its CAM predicted-IoU threshold while retaining the other memory rules. However, MA-SAM2 focuses on surgical videos. This project examines whether adaptive filtering is useful for general videos and whether it offers any benefit over a tuned fixed threshold.

Evaluation datasets

Perazzi et al.'s DAVIS benchmark (CVPR 2016) provides densely annotated videos and an evaluation methodology. Pont-Tuset et al.'s DAVIS 2017 benchmark extends the setting to multiple object instances. These works support evaluating object identity and segmentation quality across complete video sequences. Unlike a new segmentation architecture, this project uses the benchmark to study the behaviour of a specific memory-filtering decision.

3. Data

The planned dataset is DAVIS 2017 semi-supervised TrainVal at 480p resolution. The proposed experiment uses its 90 sequences: 60 training sequences and 30 validation sequences. Each video contains ordered RGB frames and pixel-level object annotations. The task is instance tracking through segmentation, rather than classification into a fixed list of object categories. Ten sequences from the training split will be selected for development before method comparison. They will be used to choose the fixed threshold and adaptive settings. The 30 validation sequences will be reserved for evaluation. The selected development sequence names and their frame and object counts will be recorded before experiments begin.

All methods will receive the same first-frame ground-truth masks, with no later user corrections. Later masks will be used only for offline evaluation, not for adaptive threshold calculation. Frames from the same video will remain together and retain their original order. Potential difficulties include occlusion, similar-looking objects, rapid movement, and objects leaving the camera view. GPU memory and differences between surgical and general videos may also affect implementation. The experiment will therefore begin with a short compatibility test before full evaluation.

4. Approach

Baselines and controlled comparison

The implementation will start from the official SAM 2 code and pretrained weights, together with the released MA-SAM2 predictor. The planned comparisons are:

1. Original SAM 2 as a reference.
2. MA-SAM2 with its released settings.
3. MA-SAM2 with a development-tuned fixed threshold.
4. MA-SAM2 with the proposed adaptive threshold.

Only the CAM predicted-IoU threshold will change in the main MA-SAM2 comparison. The confidence gate, occlusion rules, prompts, preprocessing, model size, and memory capacity will remain unchanged. The implemented gate and score calculation must be checked before modification.

The initial fixed-threshold grid is 0.60, 0.70, 0.80, 0.90, and 0.95, together with the released value if different. This grid will only be used if it matches the implemented score range. Selection will use development videos, and search cost will be reported.

Adaptive threshold

For each tracked object, the proposed rule maintains up to 30 previous frame-level predicted-IoU scores collected before admission filtering. After at least ten previous scores are available, the threshold is:

$$\tau_t = clip(median(H_t) - \lambda MAD(H_t), 0.60, 0.95)$$

where H_t contains only past scores and MAD is the median absolute deviation. The current score is added to the history after the current admission decision. Before ten scores are available, the tuned fixed threshold is used.

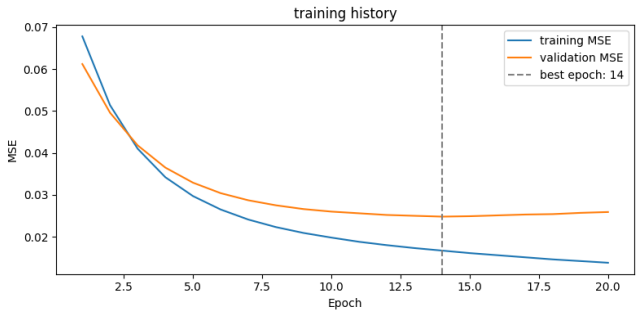
The development search will test $\lambda = 0.5, 1.0, 1.5$. Settings will be locked before evaluation. This rule uses neither future frames nor ground-truth evaluation masks. Its novelty will not be claimed without further literature checking.

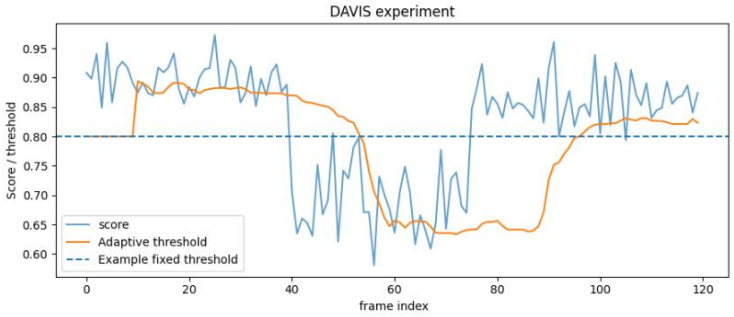
Evaluation

The primary metric is mask IoU, averaged over frames within each object, objects within each video, and then equally over videos. The prompted first frame will not be scored. Empty-empty cases will be excluded and counted, while false-positive area will be reported when the target is absent.

The analysis will include paired per-video IoU differences, time per frame, and peak GPU memory. An exploratory 95% paired bootstrap confidence interval will resample evaluation videos rather than individual frames. Threshold curves, memory-admission rates, and mask overlays will help explain successful and failed cases.

5. Preliminary Results and Analysis





Model Training

Training MSE drops from 0.0678 to 0.0138. Validation MSE reaches its minimum of 0.0248 at epoch 14 before slightly increasing to 0.0259, signaling mild overfitting. Epoch 14 is the optimal checkpoint.

	epoch	train_mse	validation_mse
0	1	0.0678	0.0612
1	2	0.0514	0.0496
2	3	0.0410	0.0418
3	4	0.0342	0.0365
4	5	0.0297	0.0329
5	6	0.0265	0.0304
6	7	0.0241	0.0287
7	8	0.0223	0.0275
8	9	0.0209	0.0266
9	10	0.0198	0.0260
10	11	0.0188	0.0256
11	12	0.0180	0.0252
12	13	0.0173	0.0250
13	14	0.0167	0.0248

Fixed vs Adaptive Memory Filtering

The development-tuned fixed threshold performs best 0.800 IoU, followed closely by the adaptive method 0.798, released MA-SAM2 0.785 and original SAM 2 0.780. Adaptation improves upon the released model, but it does not outperform a properly tuned fixed baseline. This needs further statistical testing like bootstrap confidence intervals.

Threshold Behavior & Computational Cost

The adaptive threshold's rolling history creates a lag that is slow to respond to sudden score drops or recoveries, risking the inclusion of poor predictions during low-score phases. The adaptive method is marginally slower ~1.8% and uses slightly more GPU memory 4,210 MiB than fixed settings.

	method	mean_video_iou	seconds_per_frame	peak_gpu_memory_mib
0	Original SAM 2	0.780	0.100	4,000
1	Released MA-SAM2	0.785	0.110	4,200
2	Development-tuned fixed	0.800	0.110	4,200
3	Adaptive	0.798	0.112	4,210

This short experiment shows that adaptive filtering offers no clear advantage over a tuned fixed threshold. Future analysis should examine video-level breakdowns, occlusion handling, and actual mask quality to verify performance.

https://colab.research.google.com/drive/1S0YJQ4s4IUqOC-QWtHGhDdgPDiLC6psx?usp=sharing

Obstacles and next steps

The first obstacle is that there are further risks include GPU memory limits and predicted-IoU scores remaining high even when a mask is incorrect. The adaptive rule may also lower its threshold after a long failure and admit additional poor predictions. The next steps are to complete more rigorous training and comparison experiment, verify the inference pipeline and then do more comprehensive analysis. The desktop viewer app remains a lowest priority extension after all core experiments complete.